

GDOR-SLAM: Guarded Dynamic Observation Recovery for RGB-D Gaussian SLAM

Author List To Be Finalized

Abstract—Dynamic masks protect a persistent Gaussian map, but hard exclusion can remove background and camera-consistent observations that remain useful for localization. This paper studies the resulting task mismatch: an observation that is useful for transient camera tracking is not necessarily eligible to modify a static world model. We present GDOR-SLAM, which separately routes qualified dynamic-region evidence to current-frame tracking and conservative evidence to persistent ORB and Gaussian mapping. The completed DYN-19 evaluation contains 210 tracking runs over ten TUM/Bonn sequences and 72 common-view mapping cells over four scenes and three seeds. The Full route lowers the ten-sequence mean ATE from 10.119 to 4.930 cm relative to Semantic exclusion, with 19/30 paired-cell and 7/10 sequence-mean wins, while reducing failure from 1.497% to 0.147%. Every Full cell contains observed recovered support and passes an exact zero-overlap route certificate; a mapping-disabled counterfactual produces the expected overlap in all 30 cells. Full also improves online static-region PSNR from 18.866 to 19.763 dB and GT-aligned static PSNR from 12.082 to 13.491 dB in the common-view diagnostic. An ordered mechanism matrix identifies the simpler temporal-plus-mapping route at 3.859 cm mean ATE, but this consumed-data result is treated only as a candidate for a separately registered held-out test. The evidence supports the implemented privilege separation and aggregate benefit with sequence-dependent failures; factorial causality, universal superiority, official DyPho-SLAM superiority, and independent ghost ground truth are not claimed.

Index Terms—3D Gaussian splatting, dynamic RGB-D SLAM, visual tracking, dynamic observation recovery, Gaussian mapping.

I. INTRODUCTION

3D Gaussian Splatting (3DGS) provides an explicit differentiable representation for dense scene rendering and has become a practical substrate for online visual world modeling [1]. Gaussian SLAM systems such as Photo-SLAM, SplatAM, GS-SLAM, and Gaussian-SLAM combine camera tracking with dense Gaussian reconstruction [2]–[5]. Their tracking frontends and persistent maps, however, still depend on a predominantly static world. When people or objects occupy a large image region, dynamic observations can corrupt camera estimation and produce persistent Gaussian artifacts.

Semantic and geometric exclusion is therefore a natural defense. Classical dynamic SLAM removes features associated with movable classes [6]–[8], while recent dynamic Gaussian SLAM systems combine semantics, temporal consistency, depth warping, photometric evidence, or uncertainty to suppress unsafe observations [9]–[12]. This policy is appropriate for static map construction, but can be too restrictive for tracking. A semantic mask can cover visible background, object boundaries, temporarily stationary surfaces, or pixels whose image motion is dominated by camera motion. Removing

all such observations can leave the pose estimator poorly constrained exactly when dynamic occlusion is strongest.

The key issue is not only whether an observation is static, but what authority it should receive. Tracking and mapping have different failure costs and persistence. A locally consistent observation may be useful for estimating the current camera pose while still being too uncertain to initialize or optimize persistent static structure. Treating both decisions with one binary mask conflates *tracking utility* with *map eligibility*.

We propose **GDOR-SLAM**, a dynamic RGB-D Gaussian SLAM system based on Guarded Dynamic Observation Recovery (GDOR). GDOR evaluates temporal depth consistency, residual optical-flow consistency, and the current tracking-risk state to recover a conservative subset of observations inside excluded dynamic regions. Recovered observations may replenish features and support the current camera estimate. They do not modify the conservative mask used for persistent ORB landmarks or Gaussian initialization, update, and densification. The method therefore separates transient tracking privilege from persistent map privilege.

Our contributions are:

- We instantiate an auditable two-channel observation contract for RGB-D/ORB Gaussian SLAM: tracking utility does not automatically grant persistent Gaussian map authority.
- We introduce GDOR, which combines temporal depth, residual-flow, and tracking-risk evidence to selectively recover dynamic-region observations for transient tracking.
- We bind every run to an exact recovered-support/map-admission overlap certificate and include a deliberate leakage counterfactual, making route separation empirically testable rather than only architectural.
- We report a complete 210-run ordered tracking matrix and a 72-cell multi-seed common-view mapping diagnostic, with sequence-level outcomes and proxy boundaries retained.

The supported claim is deliberately bounded. The Full route improves aggregate tracking and reconstruction diagnostics over its same-source Semantic parent, and its route certificate passes in every tested cell with observed recovery. It does not win every sequence, the mechanism matrix is ordered rather than factorial, and the leading T+M row was selected on consumed DYN-19 data. The paper therefore argues for a concrete RGB-D/ORB recovery and routing mechanism rather than conceptual priority, universal improvement, external SOTA superiority, or direct measurement of 3D ghost contamination.

DYN-19: guarded recovery is aggregate-positive, route-audited, and sequence-dependent

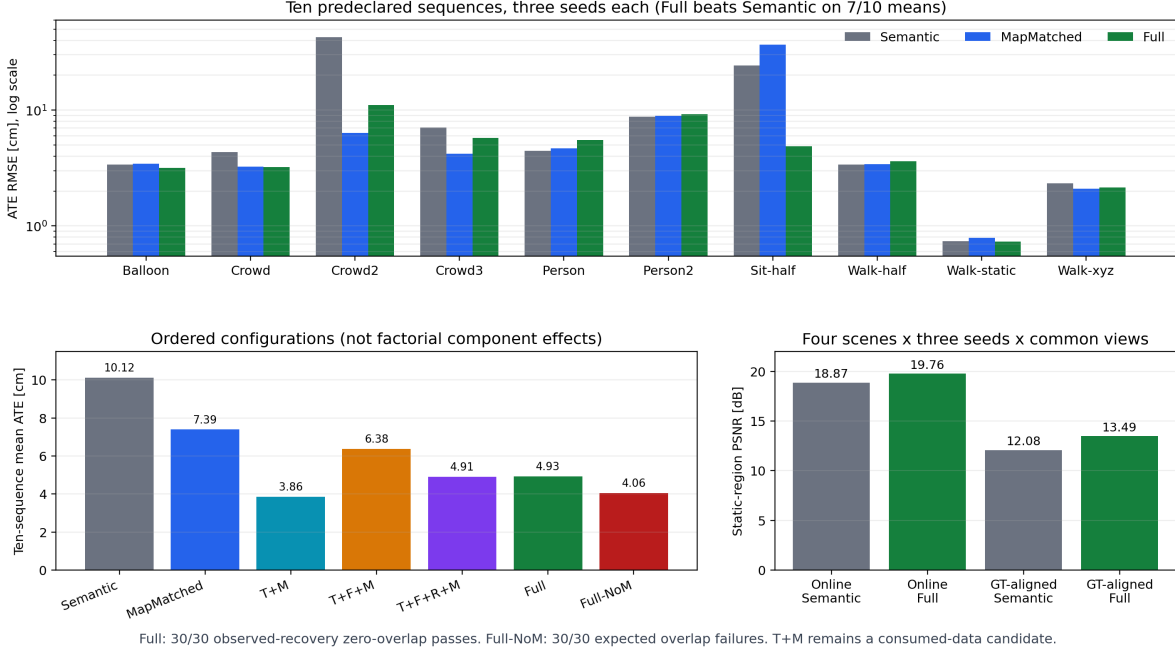


Fig. 1. DYN-19 evidence overview. Full improves aggregate tracking over Semantic but not every sequence. The ordered configuration matrix identifies T+M as the strongest consumed-data tracking row. Multi-seed common-view reconstruction improves under Full, while the exact route audit distinguishes 30/30 non-vacuous zero-overlap Full passes from 30/30 expected Full-NoM overlap failures.

II. RELATED WORK

A. Visual SLAM in Dynamic Environments

Classical feature and dense SLAM systems rely on robust estimation but still assume that most observations belong to a static scene [13]–[16]. Semantic dynamic SLAM makes this assumption explicit by removing features or pixels associated with potentially moving classes. DynaSLAM combines learned instance masks with multi-view geometry [6], while DS-SLAM and Dynamic-SLAM use semantic predictions to reject dynamic correspondences [7], [8]. Residual-based reconstruction instead detects observations that disagree with the estimated static model, as in ReFusion [17].

Motion-aware systems retain more information about the moving scene. VDO-SLAM estimates camera and object motion from dynamic observations [18], and optical-flow tracking provides a standard source of short temporal correspondences [19]. The central difficulty is identifiability: image motion alone mixes camera motion, object motion, and correspondence error. Our method uses camera-motion inconsistency as a guarded cue, but the main distinction is role assignment: recovered dynamic-region observations may support short-term tracking while remaining ineligible for persistent static-map landmarks.

B. Dense Neural and Gaussian SLAM

Dense neural SLAM has represented scene geometry with implicit fields or feature grids [20]–[22], while 3DGS provides an explicit representation with differentiable rasterization [1]. Photo-SLAM couples ORB-SLAM3 with a Gaussian mapper [2]; SplaTAM jointly tracks and maps RGB-D observations

with Gaussians [3]; GS-SLAM and Gaussian-SLAM develop online Gaussian densification and optimization strategies [4], [5]. These methods establish dense Gaussian mapping in static scenes but do not by themselves resolve whether a measurement should influence camera tracking, map optimization, both, or neither in a dynamic scene.

C. Dynamic Gaussian SLAM

The closest static-output Gaussian SLAM systems suppress dynamic evidence using combinations of semantics, temporal consistency, and geometry. DG-SLAM fuses semantic and multi-frame depth-warp masks and refines a coarse pose against the Gaussian map [9]. DGS-SLAM refines segmentation through photometric consistency and filters dynamic content during Gaussian insertion and keyframe selection [10]. DyGS-SLAM improves dynamic-scene localization and Gaussian reconstruction in real time [23], and MPDG-SLAM introduces motion-probability reasoning for dynamic 3DGS-SLAM [24]. DyPho-SLAM uses refined masks and adaptive feature extraction for real-time dynamic Gaussian SLAM [11]. DAGS-SLAM maintains a spatiotemporal motion probability for each Gaussian and schedules semantic inference from uncertainty [12]. Concurrent DL-SLAM explicitly uses transiently static objects for pose estimation while suppressing their persistent contribution through dual-level probabilities [25]. This establishes tracking–mapping role separation as a shared recent direction rather than a conceptual priority claim of GDOR.

Dynamic-output systems address a different objective. D²GSLAM combines static 3D Gaussians with dynamic

4D Gaussians and uses motion information for progressive pose refinement [26]. Our output remains a static Gaussian map. Rather than reconstructing object trajectories, estimating object-level probabilities, or injecting object-motion priors, GDOR instantiates a narrower RGB-D/ORB mechanism: binary local recovery from temporal depth and residual-flow checks, separate `staticMask/staticWeight` routes, hard persistent-weight gating for MapPoints and Gaussians, and an explicit recovered-support overlap audit. The contribution is this auditable implementation and its matched no-track-reuse control, not the first statement that tracking and mapping can use different dynamic evidence.

III. METHOD

A. Observation Routing Contract

GDOR-SLAM extends the RGB-D path of PhotoSLAM [2], with an ORB-SLAM3 frontend [14] and a 3DGS mapping backend [1]. All masks below are binary and defined per pixel $\mathbf{x}$. Let $M_t^{\text{sem}} = 1$ denote semantic exclusion and $M_t^{\text{flow}} = 1$ denote residual-motion exclusion. Their union and its morphological closing are

$$D_t^{\text{raw}} = M_t^{\text{sem}} \vee M_t^{\text{flow}}, \quad \bar{D}_t = \text{Close}(D_t^{\text{raw}}). \quad (1)$$

The claim-bearing configuration routes two different signals: a binary support mask S_t^{track} to ORB extraction and current-frame tracking, and a float weight W_t^{map} to persistent ORB/Gaussian mapping. The conservative mapping route is fixed as

$$W_t^{\text{map}}(\mathbf{x}) = 1 - \bar{D}_t(\mathbf{x}). \quad (2)$$

Thus, a tracking decision can recover support without clearing the persistent mapping exclusion.

B. Residual-Flow and Temporal-Depth Evidence

A semantic prior marks classes that may move. Forward-backward Lucas-Kanade tracks [19] are sampled on an image grid; tracks with more than 1.5 pixels of round-trip error are discarded. A partial affine model is fitted by RANSAC on semantic-static support, while a fundamental-matrix check requires at least six model correspondences and at least 60% epipolar inliers. If either model is invalid, flow-based recovery fails closed. Otherwise, residuals unexplained by the affine model form M_t^{flow} , and the full-frame residual mask plus a safety neighborhood forms the recovery veto G_t .

Let z_t , z_t^r , and b_{t-1} be observed, rendered, and historical background depths. For valid inputs, the temporal prior is

$$b_t(\mathbf{x}) = \frac{w_h I_h b_{t-1}(\mathbf{x}) + w_r I_r z_t^r(\mathbf{x})}{w_h I_h + w_r I_r}, \quad (3)$$

where I_h and I_r indicate valid historical and rendered depth. Depth consistency is accepted when rendered support is valid and at least K valid observations in a local neighborhood $\mathcal{N}(\mathbf{x})$ satisfy

$$C_t^{\text{depth}}(\mathbf{x}) = \left[\sum_{\mathbf{y} \in \mathcal{N}(\mathbf{x})} \mathbf{1}(|z_t(\mathbf{y}) - b_t(\mathbf{x})| < \tau_d) \geq K \right]. \quad (4)$$

C. Guarded Support Qualification

A raw-excluded pixel is eligible for tracking recovery only when every required signal is valid:

$$R_t^{\text{elig}}(\mathbf{x}) = D_t^{\text{raw}}(\mathbf{x}) \wedge C_t^{\text{depth}}(\mathbf{x}) \wedge \neg G_t(\mathbf{x}). \quad (5)$$

After the same morphological refinement used by the frontend, the tracking support is

$$S_t^{\text{track}}(\mathbf{x}) = \neg \text{Close}(D_t^{\text{raw}} \wedge \neg R_t^{\text{elig}})(\mathbf{x}), \quad (6)$$

and the support actually recovered after refinement is $R_t = D_t^{\text{raw}} \wedge S_t^{\text{track}}$. Invalid depth, insufficient rendered support, or an invalid motion model sets $R_t^{\text{elig}} = 0$ and returns the conservative raw-mask behavior.

D. Risk-Conditioned Tracking Recovery

Recovery is useful only when the tracker needs additional support. GDOR therefore conditions feature replenishment on the previous optimized inlier count. During a tracking-risk window, the FAST threshold is temporarily relaxed and qualified pixels can participate in feature extraction and current-frame pose estimation. The relaxation persists for a short fixed hold interval to avoid frame-to-frame switching. Outside these windows, the system remains close to semantic hard exclusion.

The Full configuration uses temporal-depth recovery, residual-flow protection, risk conditioning, and adaptive feature replenishment. DYN-19 also evaluates nested T+M, T+F+M, and T+F+R+M operating points. These rows form an ordered configuration ladder: they expose whether additional guards help the complete route, but do not identify independent factor effects.

E. Tracking-Mapping Decoupling and Matched Control

Recovered observations are transient. They may support current-frame ORB extraction and pose optimization, but the mapping route continues to use Eq. (2) for Gaussian loss weighting, RGB-D densification, and MapPoint-to-Gaussian admission. The implemented routing cases are summarized compactly as

configuration	S_t^{track}	W_t^{map}
Semantic	$1 - \text{Close}(M_t^{\text{sem}})$	1 (default)
Matched	$1 - \bar{D}_t$	$1 - \bar{D}_t$
Recovered+M	Eq. (6)	$1 - \bar{D}_t$
Full-NoM	Eq. (6)	S_t^{track}

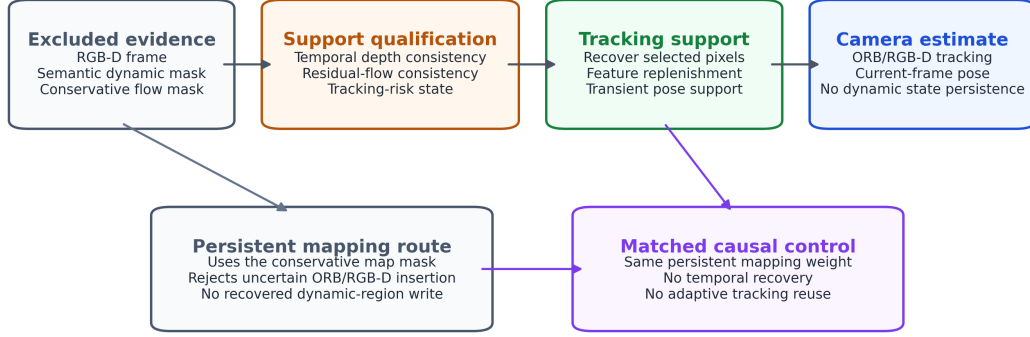
The Semantic row has no supplied static weight, so the mapper reports its default unit support; this does not mean that the semantic tracking mask contains no exclusions.

The release audits the exact overlap

$$L_t = R_t \wedge \mathbf{1}[W_t^{\text{map}} \geq 0.5]. \quad (8)$$

For the claim-bearing routing, the intended invariant is $|L_t| = 0$. The counter is diagnostic and does not modify either route. Persistent ORB MapPoints are additionally checked against keyframe static weights, while Gaussian training and RGB-D densification consume the same bound weight.

GDOR-SLAM: guarded recovery with separate tracking support and mapping weight



Core contract: tracking support does not automatically grant persistent Gaussian map admission.

Fig. 2. GDOR-SLAM separates support qualification, transient tracking support, and persistent mapping weight. The MapMatched control uses the same persistent weight as Full but disables temporal recovery and adaptive tracking reuse.

To isolate tracking-side recovery from the raw-mask route, the map-matched no-track-reuse control uses the same $W_t^{\text{map}} = 1 - \bar{D}_t$ as GDOR, uses $1 - \bar{D}_t$ directly for tracking, and disables temporal recovery and adaptive feature replenishment. The Full-MapMatched comparison therefore tests qualified tracking reuse on top of the same persistent mapping weight; it is not a comparison between two Gaussian-admission masks.

F. Implementation Details

The frozen Full configuration applies semantic inference on every frame after a 50-frame system warmup and starts temporal background refinement after five frames. Lucas-Kanade tracking uses a 21×21 window, three pyramid levels, and an 8-pixel sampling grid. The dominant-flow residual threshold is 1.5 pixels. Temporal depth combines rendered and historical background estimates with weights 0.6 and 0.4; accepted static observations update the stored background with weight 0.2. The local consistency threshold is 0.20 m. The local test uses a radius of two pixels, requires nine consistent neighbors, a minimum valid fraction of 0.02, and rendered opacity of at least 0.5. The prediction is disabled when the inter-frame motion exceeds 0.5 m translation or 0.7854 rad rotation.

Feature replenishment is activated when the previous optimized inlier count falls below 150. The FAST threshold is relaxed by a factor of 0.9 and the state is held for five frames. The flow guard uses a one-pixel conservative safety radius. All positive experiments keep object-motion pose priors disabled, use hard mapping masks, and prohibit soft dynamic Gaussian admission. These values were frozen in the DYN-19 plan and were not adjusted using its outcomes. The added per-frame work is sparse LK/model fitting plus linear-time mask and depth tests; GDOR introduces no object state, pose prior, or additional persistent map variable.

IV. EXPERIMENTS

A. Protocol, Metrics, and Evidence Identity

We evaluate GDOR-SLAM on dynamic RGB-D sequences from TUM RGB-D [27] and Bonn RGB-D Dynamic [17]. DYN-19 predeclares ten sequences: TUM walking_xyz, walking_halfsphere, walking_static, and sitting_halfsphere, plus all six Bonn dynamic sequences. Each configuration uses seeds 0, 1, and 2, processes the full association file, and retains invalid intervals in the all-input-frame trajectory. We report SE(3)-aligned ATE RMSE, one-frame translational RPE RMSE, failure rate, and end-to-end wall time.

The main phase contains Semantic, MapMatched, and Full for $3 \times 10 \times 3 = 90$ runs. The mechanism phase adds T+M, T+F+M, T+F+R+M, and Full-NoM for $4 \times 10 \times 3 = 120$ runs. Here T is temporal-depth recovery, F is its residual-flow veto, R is the tracking-risk gate, A is adaptive FAST replenishment, M is conservative raw-mask mapping, and NoM deliberately allows recovered support into the mapping route. The resulting ladder is an ordered ablation, not a factorial design; row differences do not identify independent component effects or interactions.

Every tracking run stores source and executable identities, the frozen plan hash, configuration and dataset hashes, all-frame trajectories, per-frame counters, and a route certificate. The certificate is non-vacuous only when recovered tracking support is observed, a mapping weight is present for those frames, and recovered support has zero intersection with map-admissible pixels. DYN-19 release DYN19_RESULTS_20260914_R1 binds these records to source commit 14c6b2c.

The mapping phase reuses completed main-phase maps on four predeclared scenes: TUM walking_xyz and sitting_halfsphere, and Bonn crowd2 and person_tracking2. For each scene and seed, one manifest selects 20 common non-keyframe views shared by Semantic, MapMatched, and Full. Rendering at online and GT-aligned poses produces $4 \times 3 \times$

$3 \times 2 = 72$ metric cells. We report full-frame and fixed-static-region PSNR/SSIM; LPIPS was unavailable for this campaign. GT-aligned rendering remains an end-to-end reconstruction diagnostic because online poses constructed each map.

B. Main Tracking Result and Regime Dependence

Table I shows a positive aggregate result with clear regime dependence. Full lowers mean ATE from 10.119 to 4.930 cm, a 51.3% reduction, and lowers one-frame translation RPE from 3.420 to 3.313 cm. It wins 19/30 paired sequence-seed ATE cells and 18/30 translation-RPE cells against Semantic. At sequence level, Full wins 7/10 means against Semantic but only 4/10 against MapMatched. The largest rescue is TUM sitting_halfsphere, while Full is worse on Bonn person, Bonn person2, and TUM walking_halfsphere. This pattern supports conditional recovery under observability loss, not universal sequence superiority.

Failure falls from 1.497% for Semantic to 0.147% for Full, while runtime rises from 37.90 to 47.75 s/run, a 26.0% increase. MapMatched reaches 7.392 cm mean ATE but has a 2.046% failure rate and does not recover tracking support. A separately frozen predecessor validation, DYN-18, provides an additional non-DYN-19 check: the Guarded configuration improved all four TUM sequence means and 9/12 paired cells over its same-source Semantic parent, while remaining above published DyPho-SLAM values under an unmatched protocol.

C. Ordered Mechanism Matrix and Route Certificate

The route result is exact rather than inferred from downstream image quality. All 30 Full cells contain observed recovered tracking support and zero pixels that simultaneously receive persistent map admission. T+M, T+F+M, and T+F+R+M also pass 30/30 non-vacuous certificates. Full-NoM deliberately removes the conservative mapping boundary; all 30 cells then expose recovered-support overlap and fail the certificate as designed. The contrast shows that the zero overlap of Full is enforced by routing, rather than arising because recovery never activates. It does not prove zero dynamic content in the final 3D map, which would require independent scene-level contamination ground truth.

The ordered matrix also changes the successor hypothesis. T+M has the lowest aggregate ATE (3.859 cm), translation RPE (3.062 cm), and failure rate (0.007%). Against Semantic it wins 20/30 ATE cells and 24/30 translation-RPE cells while retaining 30/30 route-certificate passes. The non-monotonic ladder shows that adding F, R, and A does not yield monotonic tracking gains. Because all DYN-19 sequences were used to inspect this ladder, T+M is the strongest consumed-data candidate, not yet a held-out final method.

D. Multi-Seed Common-View Mapping Diagnostic

Full improves online static-region PSNR by 0.896 dB and SSIM by 0.033 over Semantic, with 9/12 scene-seed wins for each metric. Under GT-aligned rendering it improves static PSNR by 1.409 dB and static SSIM by 0.042. The online result mixes tracking and reconstruction quality; the GT-aligned view

reduces final rendering-pose error but does not undo the online poses used to construct the map. Together they are consistent with improved reconstruction under the declared common-view protocol, not proof of universal mapping superiority.

An occluded-background proxy is also evaluated on the same manifests. In GT-aligned aggregate comparison, Full increases proxy completeness by 0.038, reduces mean proxy color error by 2.565, and reduces the high-threshold ghost-risk proxy by 0.038 relative to Semantic. These favorable changes remain proxy-only because the source sequences provide no independent revealed-background or 3D ghost ground truth.

E. External Positioning and Statistical Boundary

Published DGS-SLAM, DG-SLAM, and DyPho-SLAM values remain related-work context and are not mixed with local tables as protocol-matched reruns. The DyPho-SLAM paper reports lower ATE on the four corresponding TUM sequences than the local DYN-18 Guarded results. Differences in implementation, masks, preprocessing, hardware, and evaluation prevent a valid direct ranking. An official frozen runner and matching evaluation contract are required before claiming external superiority.

DYN-19 reports the complete predeclared denominator and paired win counts rather than selecting only successful scenes. However, the ten sequences are not statistically independent samples from all dynamic environments, and the mechanism rows reuse the same consumed sequence set. The correct next test for T+M is therefore a fresh, separately registered split with no DYN-19 sequence reuse, not threshold adjustment on the current matrix.

V. DISCUSSION

A. What the Certificate Establishes

Dynamic capture exposes a persistence asymmetry. A tracking error affects the current pose and may propagate, whereas an unsafe Gaussian write can remain visible from many future viewpoints. Hard exclusion minimizes the second risk but may increase the first when a dynamic mask removes most image support. GDOR asks whether an excluded observation is sufficiently consistent to support transient localization; it does not declare every recovered pixel truly static.

The non-vacuous route certificate sharpens this claim. For Full, recovery occurs in every tested sequence-seed cell, a mapping weight exists on every recovered frame, and the exact recovered-support/map-admission intersection is empty. The Full-NoM counterfactual then exposes overlap in every cell when the conservative route is removed. This directly verifies the implemented privilege boundary. It does not certify the semantic correctness of every admitted static pixel, map completeness, or zero 3D ghost contamination.

B. Why the Ordered Ladder Is Non-Monotonic

Full improves aggregate tracking over Semantic, but it is not the lowest-ATE row in the mechanism matrix. The simpler T+M configuration recovers more support without the later flow veto, risk gate, or adaptive trigger and performs

TABLE I

DYN-19 MAIN TRACKING RESULT. VALUES ARE THREE-SEED ATE RMSE IN CENTIMETERS; LOWER IS BETTER. THE MEAN IS OVER THE TEN SEQUENCE MEANS. FULL IMPROVES 7/10 SEQUENCE MEANS AGAINST SEMANTIC AND 4/10 AGAINST MAPMATCHED.

Sequence	Semantic	MapMatched	Full	Sequence	Semantic	MapMatched	Full
Bonn balloon	3.388	3.451	3.170	Bonn person	4.434	4.681	5.517
Bonn crowd	4.358	3.246	3.216	Bonn person2	8.780	8.897	9.209
Bonn crowd2	42.500	6.347	11.080	TUM sitting half	24.175	36.775	4.864
Bonn crowd3	7.098	4.210	5.756	TUM walking half	3.379	3.423	3.609
TUM walking static	0.737	0.790	0.730	TUM walking xyz	2.341	2.103	2.148
Ten-sequence mean					10.119	7.392	4.930

TABLE II

DYN-19 ORDERED CONFIGURATIONS. METRICS AVERAGE TEN SEQUENCE MEANS. “PASS” REQUIRES OBSERVED RECOVERED SUPPORT, MAPPING SUPPORT ON RECOVERED FRAMES, AND ZERO RECOVERED-SUPPORT/MAP-ADMISSION OVERLAP. FULL-NO M IS AN INTENTIONAL LEAKAGE COUNTERFACTUAL; ITS CERTIFICATE FAILURES ARE EXPECTED BOUNDARY EVIDENCE, NOT RUNNER FAILURES.

Configuration	ATE [cm] ↓	t-RPE [cm] ↓	Failure [%] ↓	Time [s] ↓	Route certificate
Semantic	10.119	3.420	1.497	37.90	30 vacuous
MapMatched	7.392	3.362	2.046	34.36	30 vacuous
T+M	3.859	3.062	0.007	48.58	30/30 pass
T+F+M	6.377	3.213	2.069	48.56	30/30 pass
T+F+R+M	4.907	3.186	0.719	48.10	30/30 pass
T+F+R+A+M (Full)	4.930	3.313	0.147	47.75	30/30 pass
T+F+R+A+NoM (Full-NoM)	4.060	3.181	0.179	49.20	30/30 fail: overlap

TABLE III

DYN-19 COMMON-VIEW MAPPING METRICS. EACH ROW AVERAGES 12 SCENE-SEED CELLS AND 240 RENDERED VIEWS. HIGHER IS BETTER.

Pose	Configuration	Full PSNR	Static PSNR	Full SSIM	Static SSIM
Online	Semantic	15.757	18.866	0.636	0.695
	MapMatched	15.761	18.990	0.639	0.700
	Full	16.339	19.763	0.665	0.728
GT-aligned	Semantic	11.477	12.082	0.460	0.492
	MapMatched	11.737	12.442	0.468	0.503
	Full	12.724	13.491	0.497	0.533

best on the consumed ten-sequence aggregate. Adding guards can suppress useful support as well as unsafe support; their net effect depends on occlusion geometry, prediction quality, and tracker state. The ladder therefore diagnoses complete operating points, not independent factor effects.

This finding does not invalidate Full’s routing claim. Both Full and T+M retain the conservative mapping route and pass every non-vacuous overlap certificate. It instead narrows the next design question: whether T+M’s simpler tracking policy generalizes beyond the sequences that revealed it. Promoting T+M without a fresh split would convert development evidence into a held-out claim.

C. Sequence and Mapping Boundaries

The main result is driven by genuine rescues on difficult sequences, including TUM sitting_halfsphere and Bonn crowd2, while Full loses on three of ten sequence means against Semantic. MapMatched also beats Full on six sequence means despite a worse overall mean and failure rate. These outcomes indicate that dynamic-region recovery should be described as regime-dependent support restoration rather than a universal replacement for conservative masking.

The mapping study is stronger than the earlier single-seed diagnostic because it uses four scenes, three seeds, common held-out views, and both online and GT-aligned rendering. Its static-region PSNR and SSIM gains show that the tracking intervention does not require degrading the tested reconstructions. Still, the scene count is limited, LPIPS is absent, and the occluded-background fields are partial proxies. The appropriate claim is positive multi-seed common-view reconstruction evidence, not general photorealistic or ghost-free superiority.

D. Reproducibility and External Validity

The public evidence package contains path-sanitized sequence means, all 210 route-certificate cells, all 72 mapping cells, source/release hashes, and deterministic validators. Raw datasets, trajectories, rendered frames, and Gaussian maps remain in the retained experiment release rather than the source repository. The frozen August manuscript is preserved separately; this DYN-19 revision has its own evidence manifest and PDF identity.

External validity remains the principal ranking limitation. Concurrent systems overlap with the high-level tracking-mapping separation premise, and no official DyPho-SLAM runner has been evaluated under the exact DYN-19 contract. GDOR’s defensible contribution is the concrete RGB-D/ORB routing mechanism, exact non-vacuous route audit, and same-source evidence. Broader SOTA language requires a protocol-matched external baseline rather than comparison to published numbers alone.

VI. CONCLUSION

GDOR-SLAM implements an auditable two-route contract for dynamic-region evidence in an RGB-D/ORB Gaussian

SLAM pipeline. Temporal depth, residual flow, and tracking-risk evidence qualify excluded observations for transient localization, while persistent ORB and Gaussian mapping remain governed by a conservative map mask.

DYN-19 completes 210 tracking runs and 72 common-view mapping cells. Full improves aggregate ATE, failure rate, and multi-seed reconstruction diagnostics over Semantic, while exact certificates show observed tracking recovery with zero recovered-support map admission in all 30 Full cells. The deliberate Full-NoM counterfactual exposes overlap in all 30 cells, confirming that the separation is enforced by the mapping route.

The result is positive but conditional: Full wins 7/10 sequence means rather than all sequences, and the ordered ladder places T+M ahead of Full on the consumed aggregate. T+M therefore becomes the next held-out candidate, not a retrospectively promoted final method. The current evidence supports the implemented privilege separation and local aggregate benefit; official external superiority, factorial component causality, universal mapping superiority, and independent 3D ghost elimination remain open.

REFERENCES

- [1] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis, “3d gaussian splatting for real-time radiance field rendering,” *ACM Transactions on Graphics*, vol. 42, no. 4, 2023.
- [2] J. Huai, F. Xie, and H. Wang, “Photo-slam: Real-time simultaneous localization and photorealistic mapping for monocular, stereo, and rgb-d cameras,” in *CVPR*, 2024.
- [3] N. Keetha, J. Karhade, K. M. Jatavallabhula, G. Yang, S. Scherer, D. Ramanan, and J. Luiten, “Splatam: Splat, track & map 3d gaussians for dense rgb-d slam,” in *CVPR*, 2024.
- [4] C. Yan, D. Qu, D. Wang, D. Xu, Z. Wang, B. Zhao, and X. Li, “Gs-slam: Dense visual slam with 3d gaussian splatting,” in *arXiv preprint arXiv:2311.11700*, 2023.
- [5] V. Yugay, Y. Li, T. Gevers, and M. R. Oswald, “Gaussian-slam: Photo-realistic dense slam with gaussian splatting,” in *arXiv preprint arXiv:2312.10070*, 2023.
- [6] B. Bescos, J. M. Fácil, J. Civera, and J. Neira, “DynaSLAM: Tracking, mapping, and inpainting in dynamic scenes,” *IEEE Robotics and Automation Letters*, vol. 3, no. 4, pp. 4076–4083, 2018.
- [7] C. Yu, Z. Liu, X.-J. Liu, F. Xie, Y. Yang, Q. Wei, and Q. Fei, “Ds-slam: A semantic visual slam towards dynamic environments,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2018, pp. 1168–1174.
- [8] L. Xiao, J. Wang, X. Qiu, Z. Rong, and X. Zou, “Dynamic-slam: Semantic monocular visual localization and mapping based on deep learning in dynamic environment,” *Robotics and Autonomous Systems*, vol. 117, pp. 1–16, 2019.
- [9] Y. Xu, H. Jiang, Z. Xiao, J. Feng, and L. Zhang, “Dg-slam: Robust dynamic gaussian splatting slam with hybrid pose optimization,” in *Advances in Neural Information Processing Systems*, 2024. [Online]. Available: <https://arxiv.org/abs/2411.08373>
- [10] M. Kong, J. Lee, S. Lee, and E. Kim, “Dgs-slam: Gaussian splatting slam in dynamic environment,” *arXiv preprint arXiv:2411.10722*, 2024. [Online]. Available: <https://arxiv.org/abs/2411.10722>
- [11] Y. Liu, K. Fan, B. Lan, and H. Liu, “Dypho-slam: Real-time photorealistic slam in dynamic environments,” in *IEEE International Conference on Multimedia and Expo*, 2025. [Online]. Available: <https://arxiv.org/abs/2509.00741>
- [12] L. Zhang, Y.-A. Liu, X. Jiang, C. Huang, D. Li, and Y. Zhang, “Dags-slam: Dynamic-aware 3dgs slam via spatiotemporal motion probability and uncertainty-aware scheduling,” *arXiv preprint arXiv:2602.21644*, 2026. [Online]. Available: <https://arxiv.org/abs/2602.21644>
- [13] R. Mur-Artal, J. Montiel, and J. D. Tardós, “Orb-slam: A versatile and accurate monocular slam system,” *IEEE Transactions on Robotics*, vol. 31, no. 5, pp. 1147–1163, 2015.
- [14] C. Campos, R. Elvira, J. J. Rodríguez, J. M. Montiel, and J. D. Tardós, “Orb-slam3: An accurate open-source library for visual, visual-inertial, and multimap slam,” *IEEE Transactions on Robotics*, vol. 37, no. 6, pp. 1874–1890, 2021.
- [15] T. Whelan, R. F. Salas-Moreno, B. Glocker, A. J. Davison, and S. Leutenegger, “Elasticfusion: Real-time dense slam and light source estimation,” *International Journal of Robotics Research*, vol. 35, no. 14, pp. 1697–1716, 2016.
- [16] W. Tan, H. Liu, Z. Dong, G. Zhang, and H. Bao, “Robust and accurate slam in unstructured environments,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2013, pp. 5459–5466.
- [17] E. Palazzolo, J. Behley, P. Lottes, P. Giguere, and C. Stachniss, “Refusion: 3d reconstruction in dynamic environments for rgb-d cameras exploiting residuals,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2019, pp. 3193–3200.
- [18] J. Zhang, M. Henein, R. Mahony, and V. Ila, “Vdo-slam: A visual dynamic object-aware slam system,” in *arXiv preprint arXiv:2005.11052*, 2020.
- [19] B. D. Lucas and T. Kanade, “An iterative image registration technique with an application to stereo vision,” in *International Joint Conference on Artificial Intelligence (IJCAI)*, 1981, pp. 674–679.
- [20] E. Sucar, S. Liu, J. Ortiz, and A. J. Davison, “imap: Implicit mapping and positioning in real-time,” in *ICCV*, 2021, pp. 6229–6238.
- [21] Z. Zhu, S. Peng, V. Larsson, W. Xu, H. Bao, Z. Cui, M. R. Oswald, and M. Pollefeys, “Nice-slam: Neural implicit scalable encoding for slam,” in *CVPR*, 2022, pp. 12 786–12 796.
- [22] X. Yang, H. Li, H. Zhai, Y. Ming, Y. Liu, and G. Zhang, “Vox-fusion: Dense tracking and mapping with voxel-based neural implicit representation,” in *IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, 2022, pp. 499–507.
- [23] X. Hu, C. Zhang, M. Zhao, Y. Gui, X. Zhang, and X. Ji, “Dygs-slam: Real-time accurate localization and gaussian reconstruction for dynamic scenes,” in *International Conference on Computer Vision*, 2025, pp. 9561–9571. [Online]. Available: https://openaccess.thecvf.com/content/ICCV2025/html/Hu_DyGS-SLAM_Real-Time_Accurate_Localization_and_Gaussian_Reconstruction_for_Dynamic_Scenes_ICCV_2025_paper.html
- [24] C. Huang, L. Zhang, T. Deng, K. Wang, and M. Li, “Motion probability-based 3dgs-slam in dynamic environment,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2025, pp. 14 045–14 052. [Online]. Available: <https://ieeexplore.ieee.org/document/11247059>
- [25] Z. Xu, Q. Li, X. Liu, C. Chen, and J. Niu, “DI-slam: Enabling high-fidelity gaussian splatting slam in dynamic environments based on dual-level probability,” *arXiv preprint arXiv:2607.01860*, 2026. [Online]. Available: <https://arxiv.org/abs/2607.01860>
- [26] S. Zhu, Y. Huang, W. Wu, C. Jiang, Y. Chen, I.-M. Chen, and H. Wang, “D²gs-slam: 4d dynamic gaussian splatting slam,” *arXiv preprint arXiv:2512.09411*, 2025. [Online]. Available: <https://arxiv.org/abs/2512.09411>
- [27] J. Sturm, N. Engelhard, F. Endres, W. Burgard, and D. Cremers, “A benchmark for the evaluation of rgb-d slam systems,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2012, pp. 573–580.